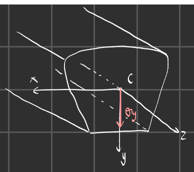
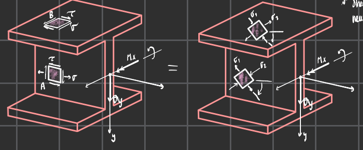

x
C
$\theta_y$
z
y

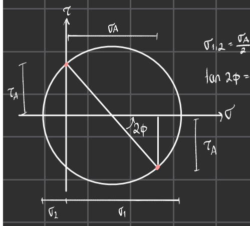
$\sigma_z = \frac{M_x}{I_x} y \qquad \frac{dM_x}{dz} = \theta_y \qquad \int_{F_1} y\,dF = S_x$

$\tau_{zy} = \frac{\theta_y S_x}{I_x b_y}$

- $\theta_y$ → kesite tesir eden ve sadece z'ye bağlı olan kesme kuvveti
- $S_x$ → taralı kesitin $(F_1)$ x eksenine göre statik momenti
- $I_x$ → tüm kesitin " " " atalet momenti
- $b_y$ → tarafsız eksenden y mesafedeki kesit genişliği

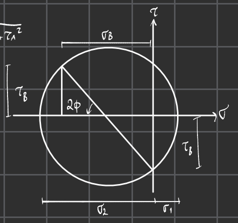
* The further away from the neutral axis, the higher the bending stress ($\sigma$).
* Shear stress is usually highest at the neutral axis and zero at the top and bottom edges.


B
τ
σ
A
τ
σ
Mx
$\theta_y$
y
=
σ₁
σ₂
σ₁
σ₂
Mx
$\theta_y$
y

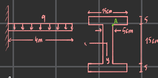

τ
σA
τA
2φ
σ
τA
σ₂
σ₁

$\sigma_{1,2} = \frac{\sigma_A}{2} \pm \sqrt{\left(\frac{\sigma_A}{2}\right)^2 + \tau_A^2}$

$\tan 2\phi = \frac{2\tau_A}{\sigma_A}$


τ
σB
τB
2φ
σ
τB
σ₂
σ₁

**Ex:** Find the max. magnitude of the distributed load q which can be safely carried by the beam,


q
4m
15cm
A
5cm
5
15
5
x
y

a) with respect to normal stresses only,

b) " " " shear " " ,

$I_x = \frac{15 \times 25^3}{12} - 2 \times \frac{5 \times 15^3}{12} = 16718.7\,cm^4$

c) " " " the combined effect of normal and shear stresses which occur at neck point A by using max. shear stress criteria.

[ $\sigma_{safety} = 14\,000\,N/cm^2$, $\tau_{safety} = 8000\,N/cm^2$, P.S.: cross-section is symmetric wrt x-y axes ]

a) $M_x = F \times r = -qL \times \frac{L}{2} = -q \times \frac{400 \times 400}{2} = -80\,000q\ (Ncm)$ $\quad \sigma_{max} = \frac{M_{max}}{I_x} y = \frac{-80\,000}{16718.7}(-25/2) \leq 14\,000 \Rightarrow q \leq 234.6\ N/cm$

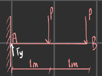
b) $S_x^G = 15 \times 5 \times (2.5 + 7.5) + 5 \times 7.5 \times (7.5/2) = 890.6\,cm^3$, $\tau_{max} = \frac{(400 \times q) \times 890.6}{5 \times (16718.7)} \leq 8000 \Rightarrow q \leq 1877.16\,N/cm$

taking the top half

c) Bending at A: $\sigma_A = \frac{-80\,000q}{16718.7}(-7.5) = 35.89q$ , $S_x^A = 15 \times 5 \times 10 = 750\,cm^3$ , $\tau_A = \frac{(400 \times q) \times 750}{5 \times (16718.7)} = 3.59q$

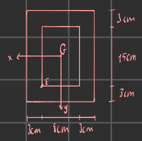
$\sigma_{1,2} = \frac{\sigma_A}{2} \pm \sqrt{\left(\frac{\sigma_A}{2}\right)^2 + \tau_A^2} = 17.94q \pm 18.3q \Rightarrow \left.\begin{array}{l} \sigma_1 = 36.4q \\ \sigma_2 = 0 \\ \sigma_3 = -.441q \end{array}\right\} \sigma_1 - \sigma_3 = 36.841q \leq 14\,000 \Rightarrow q \leq 380\,N/cm$

**Ex:** The cantilever beam is under the effect of forces P. If $\sigma_{all} = 14\,000\,N/cm^2$,


P
P
A
B
Ty
1m
1m

a) Find the max. value of P only considering the effect of bending moment

b) Draw the diagrams of normal and shear stresses at the most affected cross-section of the beam where force P is equal to the value found previously.

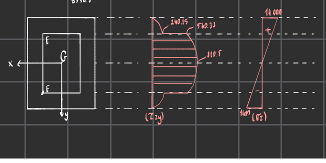

3cm
15cm
3cm
x
G
E
y
3cm
8cm
3cm

c) Check the stress at point E using the max. shear stress hypothesis.

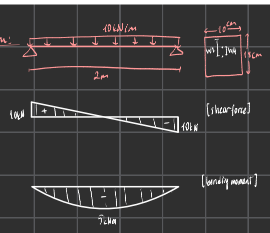
$\Sigma F_y = 0 \Rightarrow T_y = 2P$ , $\Sigma M_A = (-P) \times 1 + (-P) \times 2 = -3P$ $\quad I_x = \frac{14 \times 21^3}{12} - \frac{8 \times 15^3}{12} = 8554.5\,cm^4$

a) $\sigma_{max} = \frac{M_{max}}{I_x} y = \frac{(-3P \times 10^2)}{8554.5}(-10.5) = .368P \leq \sigma_{all} = 14\,000\,N/cm^2 \Rightarrow P \leq 38043N = 38.043\,kN$

b) $\sigma_E = \frac{(-300)(38043)}{8554.5}(7.5) \cong 10\,000\,N/cm^2$ ,


x
G
E
E
y
240.15
560.33
810.5
($\tau_{zy}$)
14 000
+
−
14000
($\sigma_z$)

**Problem:**


10kN/m
2m
10cm
18cm
10kN
+
−
10kN
[shear force]
−
5kNm
[bending moment]

[ $\sigma_{all} = 1000\,N/cm^2$; $\tau_{all} = 90\,N/cm^2$, $\theta_{max} = 10kN$; $M_{max} = 5kNm$ ]

y → represents the distance from the Neutral Axis to the point where you want to know the stress

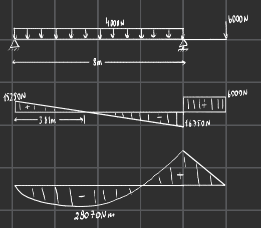
$\sigma_{max} = \frac{M_{max}}{I} y = \frac{5 \times 10^5}{10 \times 18^3/12} \cdot \left(\frac{18}{2}\right) = 930 < 1000\,N/cm^2$

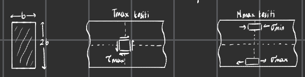
⇒ the beam is safe bending

$\tau_{max} = \frac{(10 \times 10^3) \times (10 \times 18/2) \times (18/4)}{(10 \times 18^3/12) \times \underset{\text{width}}{10}} = 83 < 90\,N/cm^2$

⇒ the beam is safe in shear

*Max. values of normal and shear stresses in a cross-section often are not in the same section and even not the same point


4000N
6000N
8m
15250N
+
3.81m
−
6000N
+
16750N
−
+
28070Nm

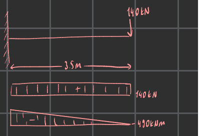

b
2b
Tmax kesiti
$\tau_{max}$
Mmax kesiti
$\sigma_{min}$
$\sigma_{max}$

✳ $\sigma_{max} = \frac{M}{\frac{bh^3}{12}} \cdot \frac{h}{2}$ [farthest fiber] $= \frac{6PL}{bh^2}$ $\qquad \tau_{max} = k_y \tau_{avg}$ [accuracy-3] $= \frac{3}{2} \cdot \frac{P}{bh}$

$\boxed{\frac{\tau_{max}}{\sigma_{max}} = \frac{h}{4l}}$

→ The cross section is dimensioned according to $\sigma_{max}$ and checked for $\tau_{max}$

**Ex:** Check whether the cross-section is sufficient. $\sigma_{all} = 14\,000\,N/cm^2$, $\tau_{all} = 8000\,N/cm^2$, $\theta_{max} = 140\,kN$, $M_{max} = 490\,kNm$

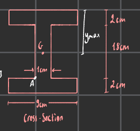

140kN
3.5m
+
140kN
−
490kNm

$I_x = \frac{1 \times 18^3}{12} + 2 \times \left(\frac{9 \times 2^3}{12} + 9 \times 2 \times 10^2\right) = 4098\,cm^4 \qquad W_x = \frac{I_x}{y_{max}} = \frac{4098}{11} = 372.5\,cm^3$

$\sigma_{max} = \frac{M_{max}}{W_x} = \frac{490kNm}{372.5cm^3} = 13151\,N/cm^2 \Rightarrow \sigma_{max} < \sigma_{all}$ : passes the bending test

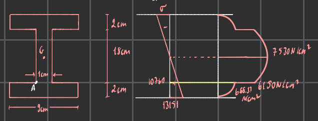
The max. shear stress is on the centre of gravity G. Static moment of lower part of the G is,

$S_x = 9 \times 2 \times (11-1) + 9 \times 1 \times (9/2) = 220.5\,cm^3 \quad \tau_{max} = \frac{\theta \cdot S_x}{b \cdot I_x} = \frac{(140 \times 10^3) \times (220.5)}{\underset{\text{width}}{1} \times 4098} = \underset{\text{safe}}{7530 < \tau_{all}}$


2cm
$y_{max}$
18cm
G
1cm
A
2cm
9cm
Cross-Section

To find the distribution of shear stress, shear stress at point A will be obtained.

$S_x = 9 \times 2 \times 10 = 180\,cm^3 \quad \tau_{A\text{-}upper} = \frac{(140 \times 10^3) \times 180}{1 \times 4098} = 6150\,N/cm^2 \quad \tau_{A\text{-}lower} = \frac{(140 \times 10^3) \times 180}{9 \times 4098} = 683.33\,N/cm^2$


G
1cm
A
9cm
2cm
18cm
2cm
σ
−
10760
13151
7530N/cm²
683.33 N/cm²
6150N/cm²

the principal stresses at point A,

$\sigma_{1,2} = \frac{\sigma_A}{2} \pm \sqrt{\left(\frac{\sigma_A}{2}\right)^2 + \tau_A^2} = \frac{10760}{2} \pm \sqrt{\left(\frac{10760}{2}\right)^2 + 6150^2}$

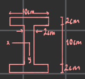
$\sigma_1 = 13550\,N/cm^2$ , $\sigma_2 = -2790\,N/cm^2$

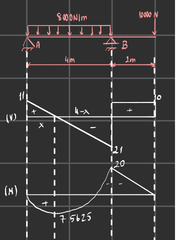
**2nd Part:** Max. Shear Stress Hypothesis (TRESCA): $\sigma_1 - \sigma_2 = \sigma_m \rightarrow 13550 - (-2790) = 16340\,N/cm^2 > 14000\ (\sigma_{all})$

⇒ The cross-section is not sufficient

**3rd Part:** Distortional Strain Energy Hypothesis (Von Mises): $\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - \sigma_1\sigma_2 - \sigma_2\sigma_3 - \sigma_3\sigma_1 \leq \sigma_m^2 \Rightarrow 15139 > 14000 \Rightarrow$ unsafe

**Ex:** For the overhanging beam, $\tau_{all} = 9000\,N/cm^2$, $\sigma_{all} = 14\,000\,N/cm^2$. Check the safety of the beam wrt shear and normal stresses. Draw the shear and normal stress distributions over the cross-section for the critical locations.


10cm
2cm
2cm
10cm
2cm
x
y


8000N/m
10000 N
A
B
4m
2m
11
+
4−x
x
−
+
10
21
(V)
20
(M)
+
−
7.5625

first we have to find the reaction forces at A(pin) and B(roller).

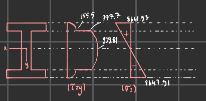
$\Sigma M_A = 0 \rightarrow B_y(4) - 8.4(2) - 10(6) = 0 \Rightarrow B_y = 31kN \quad A_y = 11kN$

A-B : $V(x) = 11 - 8x = 0 \Rightarrow x = 1.375m$ , $A = M_{local\text{-}max} = \frac{1}{2}(1.375)(11) = 7.5625\,kNm$

$T_{max} = 21kN$ , $M_{max} = 20\,kNm \qquad I_x = \frac{10 \times 14^3}{12} - 2 \times \frac{4 \times 10^3}{12} = 1620\,cm^4$

$\sigma_{max} = \frac{M_{max}}{I_x} y = \frac{20 \times 10^5}{1620} \times 7 = 8641.97\,N/cm^2 \leq 14\,000 \rightarrow$ safe ✓

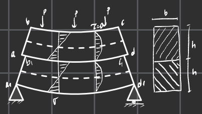
$\tau_{max} = \frac{(V_{max})(S_x)}{(I_x) \times b} = \frac{(21 \times 10^3) \times (10 \times 2 \times 6 + 2 \times 5 \times 2.5)}{(1620) \times 2} = 939.81\,N/cm^2 \leq 9000 \rightarrow$ safe ✓

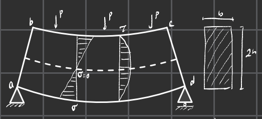

x
y
155.5
777.7
8641.97
939.81
+
−
8641.97
($\tau_{zy}$)
($\sigma_z$)

$\tau_{\substack{bottom\\flange}} = \frac{(V_{max}) \times (S_x)}{(I_x) \times b} = \frac{(21 \times 10^3) \times (10 \times 2 \times 6)}{(1620) \times (10)} = 155.55\,N/cm^2$

$\tau_{\substack{upper\\web}} = \frac{(V_{max}) \times (S_x)}{(I_x) \times b} = \frac{(21 \times 10^3) \times (10 \times 2 \times 6)}{(1620) \times (2)} = 777.77\,N/cm^2$

# Composite Beams:





→ no connection between the beams, free relative sliding - no shear occurs

$M_B = 2\left(\frac{\sigma_{all}\times(bh^3/12)}{h/2}\right) = \frac{1}{3}\sigma_{all}bh^2$

$W_{total} = 2W_1$ , $W_1 = bh^2/6$

total moment

→ connected beams (work together) - shear occurs at the interface.

$M_C = \sigma_{all}\left(\frac{b(2h)^3/12}{h}\right) = \frac{2}{3}\sigma_{all}bh^2$ $\boxed{M_C = 2M_B}$

compound beam can carry two fold load.

$W_{compound} = 4W_1$

→ relative movement will arise between ad and b1c1 edges

→ there will be no relative movement between the edges ad and b1c1

→ the sign of stress σ will change only once.

Shear force between two sections: $Z = \frac{S_x\cdot(T_y)_m}{I_x}\cdot a = Z_1 . a = q_m a$ , $Z_1$ or $q$ is shear force per unit length : shear flow

S.m. of added part

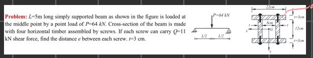
average shear stress

total m.i. of the whole section

distance between the planes

→ $S_x = b\times h\times\frac{h}{2}$ , $I_x = b(2h)^3/12 = \frac{2bh^3}{3}$

for the rectangular cross-section above



**Problem:** *L*=5m long simply supported beam as shown in the figure is loaded at the middle point by a point load of *P*=64 *kN*. Cross-section of the beam is made with four horizontal timber assembled by screws. If each screw can carry *Q*=11 kN shear force, find the distance *e* between each screw. *t*=3 cm.

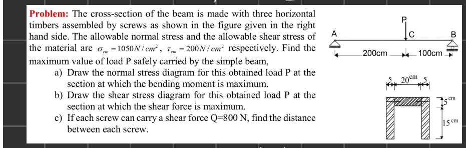
P=64 kN
L/2
L/2
22cm
area
t=3cm
8cm
t
t
G
x
12cm
t=3cm
y

$T_{max} = P/2 = 32kN$

$I_x = \frac{22\times18^3}{12} - \frac{16\times12^3}{12} = 8388cm^4$

$q = \frac{S_x\,T_y}{I_x} = \frac{(22\times3\times7.5)\times(32\times10^3)}{8388} = 1888.41N/cm$

$q.e \le 2\theta \Rightarrow e \le 11.65cm$



**Problem:** The cross-section of the beam is made with three horizontal timbers assembled by screws as shown in the figure given in the right hand side. The allowable normal stress and the allowable shear stress of the material are $\sigma_{em} = 1050N/cm^2$, $\tau_{em} = 200N/cm^2$ respectively. Find the maximum value of load P safely carried by the simple beam,

a) Draw the normal stress diagram for this obtained load P at the section at which the bending moment is maximum.

b) Draw the shear stress diagram for this obtained load P at the section at which the shear force is maximum.

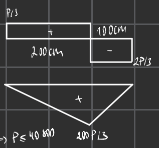
c) If each screw can carry a shear force Q=800 N, find the distance between each screw.

$\Sigma M_A = 0 \Rightarrow 2P = 3B$
$\Sigma F_y = 0 \Rightarrow A + B = P$ $\Big\}\ B = 2A$

|  | y | A | yA |
|---|---|---|---|
| 1 | 10 | 100 | 1000 |
| 2 | 17.5 | 100 | 1750 |
| 3 | 10 | 100 | 1000 |
|  |  | 300 | 3750 |

$\bar{y} = \frac{\Sigma yA}{\Sigma A} = 12.5cm$

$I_x = 2\times\frac{5\times20^3}{12} + \frac{29\times5^3}{12} + 2\times100(12.5-10)^2 + 100\times(17.5-12.5)^2 = 10625cm^4$



$\sigma_{max} = \frac{M_{max}}{I_x}\times y = \frac{200P/3}{10625}\times(12.5) \le 1050N/cm^2 \Rightarrow P \le 13387.5N$

$\tau_{max} = \frac{(V_{max})\times(S_x)}{(b)\times(I_x)} = \frac{(2P/3)\times(2\times5\times7.5\times7.5/2 + 29\times5\times5)}{(2\times5)\times(10625)} \le 200N/cm^2 \Rightarrow P \le 40\,800$

$q e \le N\theta \Rightarrow \tau b e \le N\theta \Rightarrow e \le \frac{N\theta}{\tau b} \Rightarrow e \le 3.809\ cm$

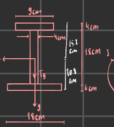
Ex: The pieces are connected by glue. For glue $\tau_{all} = 150N/cm^2$, for material $\tau_{all} = 200N/cm^2$

a) Find $T_y$ safely carried

b) Draw the shear stress diagram

$\bar{y} = \frac{2\times(18\times4) + (13\times4\times18) + (24\times9\times4)}{(18\times4) + (4\times18) + (9\times4)} = \frac{1944}{180} = 10.8cm$

$I_x = \frac{18\times4^3}{12} + \frac{4\times18^3}{12} + \frac{9\times4^3}{12} + 72(10.8-2)^2 + 72(10.8-13)^2 + 36(10.8-24)^2 = 14284.8\ cm^4$



for material, $\tau_{max} = \frac{(T_y)\times(9\times4\times13.2 + 4\times11.2\times5.6)}{(14284.8)\times9} \le 200N/cm^2 \Rightarrow T_y \le 35412.96N$

for glue, $\tau_{max} = \frac{(T_y)\times(9\times4\times13.2)}{(14284.8)\times4} \le 150N/cm^2 \Rightarrow T_y \le 18036.36\ N$

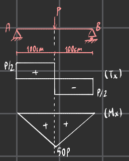
Ex: Beam with a span length of 2m is built from 3 equal timber parts (each having rectangular cross-section of 6×9 cm) by means of gluing. If for timber $\tau_{all} = 90N/cm^2$, $\sigma_{all} = 1300N/cm^2$ and for glue $\tau_{all} = 40N/cm^2$. find the greatest P load which can be carried safely

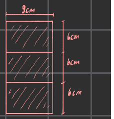
1- $I_x = \frac{bh^3}{12} = \frac{9\times18^3}{12} = 4374cm^4$ 2- $S_x^G = 9\times9\times\frac{9}{2} = 405cm^3$ 3- $S_{x-glue} = 9\times6\times6 = 324cm^3$

→ with respect to bending: $\sigma_z = \frac{50P}{4374}\,9 \le \sigma_{all} \rightarrow P \le 12636N$

→ 〃 〃 〃 shear in timber: $\tau_{zy} = \frac{V\cdot S_x}{b\cdot I_x} = \frac{(P/2)\times(405)}{9\times4374} \le \tau_{all} \rightarrow P \le 19440N$

→ 〃 〃 〃 〃 〃 glue: $\tau_{glue} = \frac{(P/2)\times(324)}{9\times4374} \le \tau_{glue} \rightarrow P \le 9720N$





Ex: Each screw can sustain 1kN of shear force. Taking into account bending with shear effects, determine "e" (distance) if P=6kN is applied to the cross-section.

$I_x = \frac{26\times40^3}{12} - \frac{20\times30^3}{12} = 93666.67cm^4$ area of top flange: $26\times5 = 130cm^2$ $\bar{y} = 20-2.5 = 17.5\ cm$ $S_x = 26\times5\times17.5 = 2275cm^3$

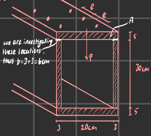
$\tau = \frac{(V)\times(S_x)}{(I_x)\times(b)} = \frac{6\times2275}{93666.67\times6} = .0243kN/cm^2$ , (shear force per unit length - shear flow) $q = \tau\times b = (.0243\ kN/cm^2)\times(6cm)$

must be resisted by the screws

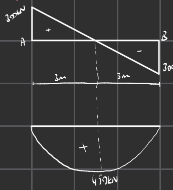
$q = .1458\ kN/cm$

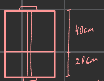
force on a segment of length "e": $q\times e$

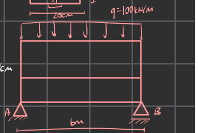
each row contains 2 screws with each having a capacity of 1kN → a pair can carry 2kN $\Big\}$ $q\times e = 2\times1$ ; $(.1458)\times e = 2 \Rightarrow e = \boxed{13.89cm}$



Ex: for bolts $\tau_{all} = 11000N/cm^2$. Find the necessary number of bolts and distance between them. Diameter of bolts is 2cm.







$P = 100\times6 = 600kN$ $I_x = \frac{bh^3}{12} = \frac{20\times60^3}{12} = 36\times10^4\ cm^4$

$q = \frac{(V)\times(S_x)}{I_x} = \frac{(300\times10^3N)\times(20\times20\times20)}{36\times10^4} = 6666.67N/cm$

$q.e \le (\pi r^2)\times\tau_{all}$

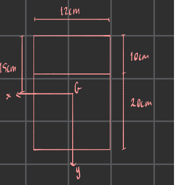
$e \le \frac{\pi r^2}{e}\times\tau_{all} \Rightarrow e \le 5.18cm$

we pick 5 cm ⇒ $\frac{600cm}{5} = 120$ bolts

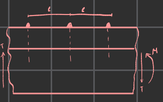
Ex: For the material $\tau_{all} = 200N/cm^2$, for the glue $\tau_{all} = 160N/cm^2$. Determine shear force $T_y$.

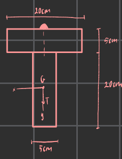
$I_x = \frac{12\times30^3}{12} = 27\,000\ cm^4$ $\tau_{max} = \frac{(V_{max})\times(S_x)}{(b)\times(I_x)}$ for glue, $\tau_{max} = \frac{(T_y)\times(12\times10\times10)}{(12)\times(27\,000)} \le 160 \Rightarrow T_y \le 43\,200$

for material, $\tau^G_{max} = \frac{(T_y)\times(12\times15\times15/2)}{(12)\times(27\,000)} \le 200 \Rightarrow T_y \le 48\,000$



Ex: A wooden beam carrying a bending moment and a shear force T=400N is made of using 2 20×5 cm planks connected by nails. Each nail can carry θ=800N. e=?





$\bar{y} = \frac{(22.5\times100) + (10\times100)}{200} = 16.25cm$

$I_x = \frac{20\times5^3}{12} + \frac{5\times20^3}{12} + 100\times(6.25)^2 + 100\times(6.25)^2 = 11\,354.167\ cm^4$

$\tau\times b = \frac{400\times(20\times5\times6.25)}{11354.167} = 22.018 \Rightarrow 22.018\times e \le 800 \Rightarrow e = 36.3cm$

$q\cdot e \le N\theta \rightarrow e \le \frac{N\theta}{\tau b}$